



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/387427

Date:

25/08/2022(17:39:48)

Subject: Approval of New LMI "THERMOGRAPHIC SURVEY OF BOILER FURNACE WALL & SCW TUBES"

OPERATION & MAINTENANCE

MECHANICAL MAINTENANCE

(PRESSURE PARTS)

REF.: SMPP/O&M/MM/PP/

Date: 25.08.2022

Subject: Approval of New LMI "THERMOGRAPHIC SURVEY OF BOILER FURNACE WALL & SCW TUBES"

LMI No: COS-ISO-00-OGN/OPS/SYST/055

This is a new LMI which was prepared as per the COS-ISO-00-OGN/OPS/SYST/055 guidelines.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

Rahul Jain - 013328 (SENIOR MANAGER ,MECH MAINT- BOILER) ,Simhadri
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On: 25/08/2022(17:39:48) **Ref:** SMTTP/MECH MAINT-/2022-23/LMI/387427

This new LMI "Thermographic survey of Boiler furnace wall & SCW tubes" has been prepared following guidelines related to the document COS-ISO--00-OGN/OPS/SYST/055.

This new LMI with revision no-0 may please be reviewed and approved by the competent authority.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

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On:26/08/2022(09:52:29) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/387427

This is a new LMI prepared based on recently issued OGN COS-ISO-00-OGN/OPS/SYST/055.

LMI checked and submitted for approval

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:27/08/2022(09:19:41) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/387427

Forwarded for further approval

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

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On:30/08/2022(16:19:03) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/387427

Submitted for approval pls

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
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On:01/09/2022(09:23:18) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/387427

New LMI no. LMI/OGN/OPS/MECH/055, **ISSUE-0; REV.-0; AUG'2022** for "THERMOGRAPHIC SURVEY OF BOILER FURNACE WALL & SCW TUBES " for NTPC Simhadri 500 MW Boilers, may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
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On:04/09/2022(12:31:27) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/387427

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

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On:05/09/2022(13:15:42) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/387427

Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

Girish Chandra Choukse - 002907 (GENERAL MANAGER ,HEAD OF PROJECT)
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On:05/09/2022(15:22:26) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/387427

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/055

ISSUE NO.0; REVISION NO. 0; DATE: AUG'2022

**THERMOGRAPHIC SURVEY OF
BOILER FURNACE WALL & SCW
TUBES**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	THERMOGRAPHIC SURVER OF BOILER FURNACE WALL & SCW TUBES
<u>2.a) DOCUMENT CODE</u> b) REVISION/ISSUE NO.&DATE	LMI/OGN/OPS/SYST/055 <u>ISSUE-0; REV.-0; AUG'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/SYST/055</u>
<u>4.PREPARED BY: SENIOR MANAGER</u> (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM(MM-BMD) AGM (MM-BMD) GM (MTC.) GM (O&M)	SIGNATURE----- SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE:----- HOP (SIMHADRI)
<u>7. DATE OF ISSUE</u>	
<u>8. INDEX LIST</u>	ENCLOSED.
<u>9. DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: THERMOGRAPHIC SURVER OF BOILER FURNACE WALL & SCW TUBES		
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<u>2.0 INTRODUCTION</u> A thermography survey is performed to inspect the external surface of the boiler tube wall, to identify and quantify possible hot spots/flow restrictions. It is one of the potent tool to detect partial to full restrictions in water/steam flow path of boiler tubes by monitoring tube experiencing high temperature, identifying it for cleaning during next opportunity shutdown and thereby avoiding boiler tube failures due to overheating. The thermography scan provides a 'picture' of surface temperature distribution across the scanned area with particular attention being paid to the areas around butt welds, bends and spiral to vertical wall forge fittings where potential debris may collect and cause blockage. The objective of this LMI is to establish a system for thermographic survey of boiler furnace wall & SCW tubes of the boiler. <u>3.0 SCOPE</u> The scope of this LMI is thermography survey (partial/complete) of the boiler furnace wall & SCW tubes of boiler to know flow restriction in boiler tubes. <u>4.0 REFERENCE DOCUMENT</u> Operation information note COS-ISO-00-OGN/OPS/SYST/055 circulated by operation services-NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> Not applicable <u>6.0 SPECIFIC INSTRUCTIONS</u> The objective of this LMI is sought to be achieved by a) Ensuring availability of thermographic imaging camera with infrared detection capability along with its calibration certificates. b) Examination and survey of the area to be done along with safe route for escape in case of any mishappening with arrangement for fire prevention and safety personnel.		

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<u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u> a) The boiler must be in startup condition for any significant obstruction. Responsibility – HOD(Operation) b) 150 mm high band around four walls – insulation and lagging should be removed to measure individual tubes temperature as per attached drawing. Responsibility - Mech. Mtc. (Pr.Parts) <u>8. EXCEPTION REPORTING</u> The exceptions encountered during inspection are to be reported as per the following format ----- Mech. Mtc.(Pr.Parts) <table border="1"> <thead> <tr> <th>Description of exception</th> <th>Unit No.</th> <th>Location of exception</th> <th>Remarks</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table> <u>9.0 RESPONSIBILITIES</u> The responsibility of carrying out different tasks in relation to thermographic survey of boiler furnace wall & SCW tubes as detailed in the LMI is with HOS (Pressure Parts)			Description of exception	Unit No.	Location of exception	Remarks				
Description of exception	Unit No.	Location of exception	Remarks							

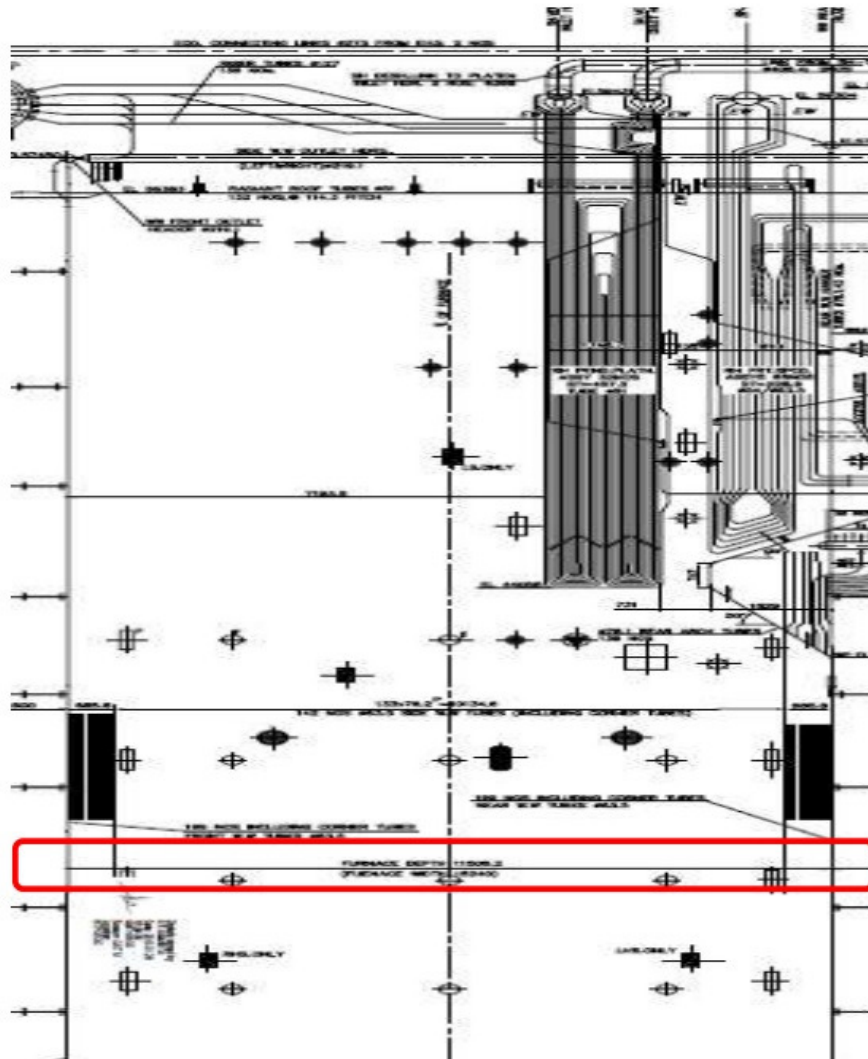
NTPC	LMI	SIMHADRI
TITLE: THERMOGRAPHIC SURVER OF BOILER FURNACE WALL & SCW TUBES REV:0 DATE: AUG'2022		
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<u>10. RECORD KEEPING</u> The records to be maintained under this LMI include a) Compliance record of the survey and mtc. activities detailed under this LMI b) Exception reporting The above records are to be maintained by Area Engr. Pr. Parts Group. <u>11.REVIEW</u> This document shall be reviewed by HOP (SIMHADRI) on TWO yearly basis.		

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[illegible]

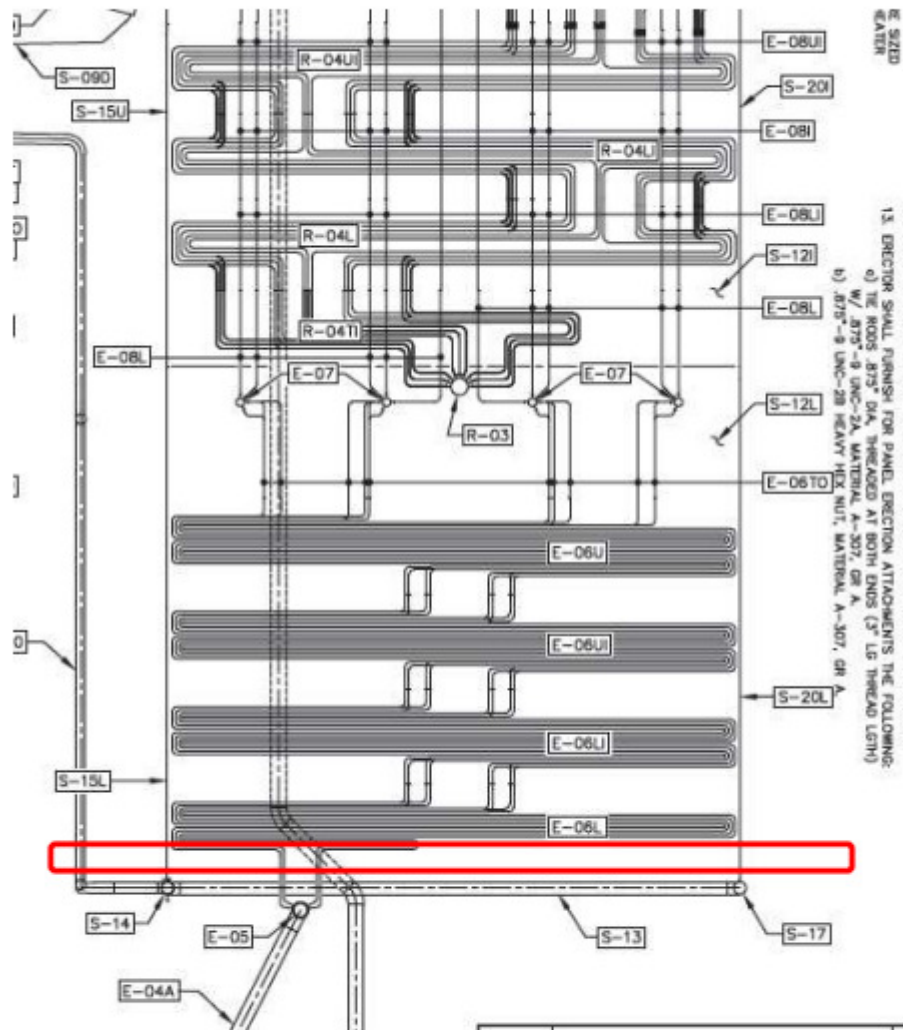
TITLE: THERMOGRAPHIC SURVER OF BOILER FURNACE WALL & SCW TUBES**REV:0****DATE: AUG'2022****REF NO. COS-ISO-00-OGN/OPS/SYST/055****PAGE : 7 OF 10**

SYSTEM FOR THERMOGRAPHIC SURVEY OF BOILER FURNACE WALL & SCW TUBES AT POWER STATIONS



Area to be checked in Sub critical Water Wall
(Above Burner)

SYSTEM FOR THERMOGRAPHIC SURVEY OF BOILER FURNACE WALL & SCW TUBES AT POWER STATIONS



Area to be checked in SCW Second Pass
(Above SCW Header)

SYSTEM FOR THERMOGRAPHIC SURVEY OF BOILER FURNACE WALL & SCW TUBES AT POWER STATIONS



Window for Thermal Scanning (after removing insulation strip)



Window for Thermal Scanning covered by Removable Insulation Boxes



Final Cladding sheet after clamping of removable strips

TITLE: THERMOGRAPHIC SURVER OF BOILER FURNACE WALL & SCW TUBES

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SYSTEM FOR THERMOGRAPHIC SURVEY OF BOILER FURNACE WALL & SCW TUBES AT POWER STATIONS

Annexure A

Date of Investigation _____
Unit No. : _____
Load : _____
Main Steam Pressure : _____
Main Steam Temperature : _____
Area to be investigated : _____
Elevation of Boiler : _____
Average Tube Temperature : _____
Frequency Of Survey : _____

S.No.	Location of Panel	Tube No.	Temperature of Crown of Tube	Remarks

Signature Of FQA

Signature Of Operation

Signature Of BMD